

SV9 - Digital Verification using SV & UVM

=====Assignment (5)=====

Applying UVM

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Part #1: Setting up till Environment

-run phase of test results:

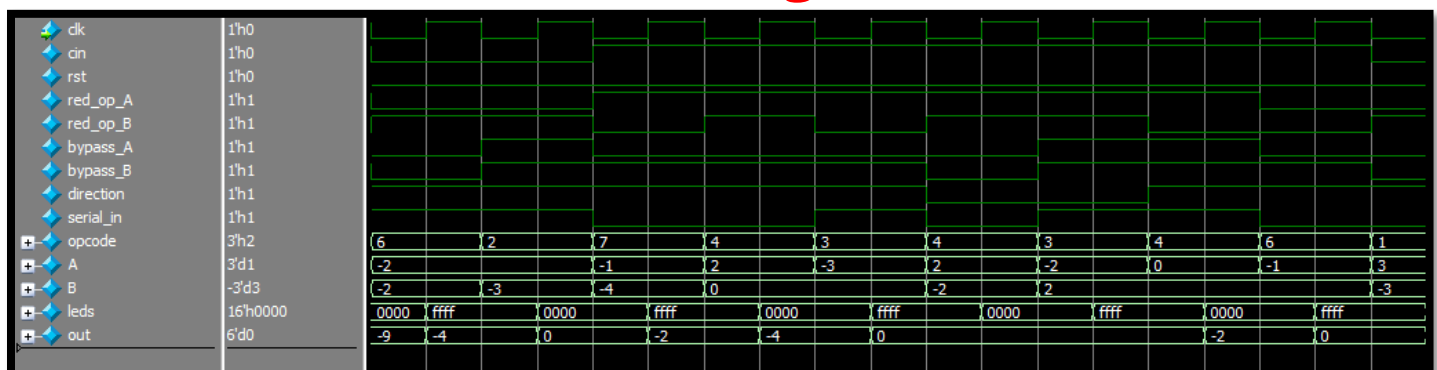
```
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(215) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(217) @ 0: reporter [Questa UVM]  questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test ALSU_test...
# UVM_INFO ../../VS_code/Session5/Assignment5/Part_1/ALSU_test.sv(47) @ 0: uvm_test_top [run_phase_of_test] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1268) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 5
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [Questa UVM] 2
# [RNTST] 1
# [TEST_DONE] 1
# [run_phase_of_test] 1
# ** Note: $finish : F:/IEEE_Digital/Installation/questasim64_10.6c/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 100 ns Iteration: 54 Instance: /top
```

Part #2: Setting up till Driver

-run phase of test results:

```
#
# (Specify +UVM_NO_RELNOTES to turn off this notice)
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(215) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(217) @ 0: reporter [Questa UVM]  questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test ALSU_test...
# UVM_INFO ../../VS_code/Session5/Assignment5/Part_2/ALSU_test.sv(46) @ 0: uvm_test_top [run_phase_of_test] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1268) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 5
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
```

-Waveform of Interface signals:

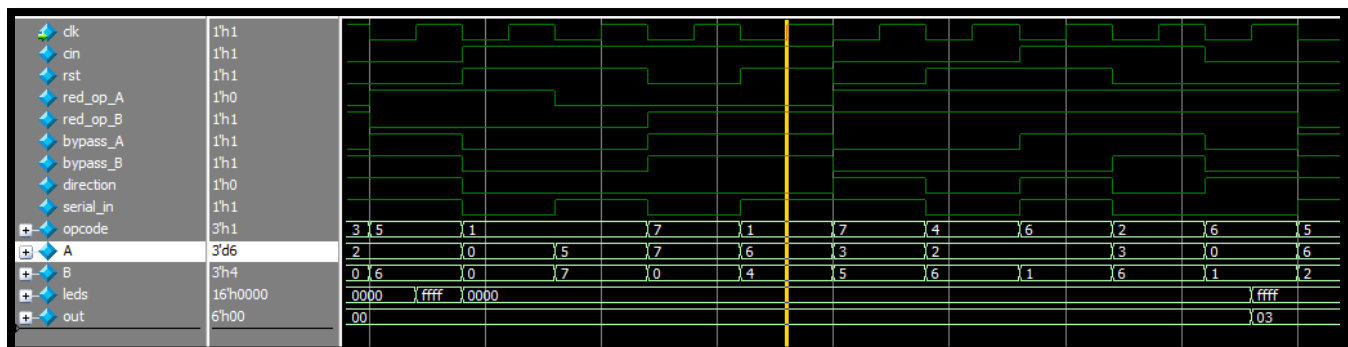


Part #3: Setting up till Driver & Config object

-run phase of test results:

```
# (Specify +UVM_NO_RELNOTES to turn off this notice)
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(215) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(217) @ 0: reporter [Questa UVM]  questa_uvm::init(all)
# UVM_INFO @ 0: reporter [RNTST] Running test ALSU_test...
# UVM_INFO ../../../../VS_code/Session5/Assignment5/Part_3/ALSU_test.sv(47) @ 0: uvm_test_top [run_phase_of_test] Inside the ALSU test
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1268) @ 100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    5
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
```

-Waveform of Interface signals:



Drive link with code:

https://drive.google.com/drive/folders/1NpWocnzCf5ofBGgY9z8PYzdz_6biPjYQ?usp=drive_link